

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	29-06-2026
Team ID	
Project Name	Nutrition Assistant
Maximum Marks	4 Marks

Functional Requirements:

Following are the Functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Registration Form (Name, Email, Password)
		Email Validation
		Store user details in MongoDB
FR-2	User Authentication	User Login using Email & Password
		JWT-based Authentication
		Secure Logout
FR-3	Dashboard Management	Display User Dashboard
		Display Daily Nutrition Summary
		Display Calories Consumed
		Display Macronutrient Distribution
FR-4	Food Search	Search food items by name
		Retrieve nutritional information
		Display calories and nutrients
FR-5	Calorie Tracking	Track daily calorie intake
		Update calorie count after food selection
		Display remaining calorie goal
FR-6	Macronutrient Analysis	Display Protein Intake
		Display Carbohydrate Intake

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		Display Fat Intake
FR-7	User Profile	View Profile
		Update User Information
FR-8	Database Management	Store User Data
		Store Nutrition Data
		Retrieve Nutrition History

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should provide a simple, intuitive, and responsive user interface for users of all experience levels.
NFR-2	Security	User passwords should be encrypted, authentication should use JWT, and only authorized users can access protected resources.
NFR-3	Reliability	The system should consistently provide accurate nutrition data and remain operational without failures.
NFR-4	Performance	Food search, dashboard loading, and API responses should complete within 2–3 seconds under normal usage.
NFR-5	Availability	The application should be accessible 24×7 with minimal downtime using cloud deployment (Vercel & Render).
NFR-6	Scalability	The system should support increasing numbers of users and nutrition records without significant performance degradation.
NFR-7	Maintainability	The MERN architecture should allow easy maintenance, debugging, and feature enhancements.
NFR-8	Compatibility	The application should function correctly on modern web browsers and different screen sizes (desktop, tablet, mobile).

NFR No.	Non-Functional Requirement	Description
NFR-9	Portability	The application should be deployable on different cloud platforms with minimal configuration changes.